

HOURLY POWER CONSUMPTION FORECASTING FOR A SMART HOME USING TIME SERIES MODELS

A PROJECT REPORT

Mini Project (01CE0609)

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CERTIFICATE

This is to certify that the project report submitted along with the project entitled “**Hourly Power Consumption Forecasting Using Time Series Models**” has been carried out by **Sharafathusein Kothariya** (92301703020), **Shubhra Gupta** (92301703015), **Anuska Singh** (92301703037) under my guidance in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering, 6th Semester of Marwadi University, Rajkot during the academic year 2025-26.

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DECLARATION

We hereby declare that the **Mini Project (01CE0609)** report submitted along with the Project entitled “**Hourly Power Consumption Forecasting Using Time Series Models**” submitted in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering to Marwadi University, Rajkot, is a bonafide record of original project work carried out by us at Marwadi University under the supervision of **Prof. Ronak Doshi** and that no part of this report has been directly copied from any students’ reports or taken from any other source, without providing due reference.

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Abstract

The rapid growth of energy forecasting technologies has intensified the need for accurate and efficient power consumption management. Electricity demand varies significantly throughout the day as a result of usage patterns and environmental conditions. This project presents a systematic comparative study of five time series forecasting models applied to the publicly available New York Independent System Operator (NYISO) hourly electricity load dataset. The models evaluated are Long Short-Term Memory (LSTM), Gradient Boosting Machine (GBM), Gated Recurrent Unit (GRU), a Hybrid LSTM+GBM ensemble, and Random Forest (RF). The standard evaluation metrics used are Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). The findings demonstrate that the Hybrid LSTM+GBM model achieves the best performance across all metrics: RMSE of 52.50 MW, MAE of 35.43 MW, MAPE of 0.60%, and R^2 of 0.9979, representing a 40.3% improvement in RMSE over the GRU baseline. The research confirms that combining the sequential temporal learning capability of LSTM with the residual error correction strength of GBM yields a highly accurate and practically deployable forecaster for intelligent energy grid management systems.

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Abbreviations

LSTM	Long Short-Term Memory
GBM	Gradient Boosting Machine
GRU	Gated Recurrent Unit
RF	Random Forest
NYISO	New York Independent System Operator
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
R²	Coefficient of Determination
MWh	Megawatt-hour
MW	Megawatt
ARIMA	Autoregressive Integrated Moving Average
SARIMA	Seasonal Autoregressive Integrated Moving Average
SVM	Support Vector Machine
RNN	Recurrent Neural Network
ACF	Autocorrelation Function
PACF	Partial Autocorrelation Function
EDA	Exploratory Data Analysis
DSM	Demand-Side Management
GPU	Graphics Processing Unit
MSE	Mean Squared Error
CAISO	California Independent System Operator
ERCOT	Electric Reliability Council of Texas
CNN	Convolutional Neural Network
IoT	Internet of Things

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CHAPTER 1

INTRODUCTION

1.1 PROJECT SUMMARY

The increasing demand for electricity across modern power grids has fundamentally changed the landscape of energy consumption management. Due to the proliferation of connected infrastructure, distributed energy systems, and renewable energy sources, the volume and complexity of hourly power consumption data have grown substantially. Accurate prediction of hourly electricity demand has emerged as a critical requirement for efficient grid operation, cost management, and optimisation of renewable energy integration.

This project, titled “Hourly Power Consumption Forecasting Using Time Series Models,” undertakes a systematic comparative evaluation of five forecasting models applied to the publicly available NYISO hourly electricity load dataset: Long Short-Term Memory (LSTM), Gradient Boosting Machine (GBM), Gated Recurrent Unit (GRU), a Hybrid LSTM+GBM ensemble, and Random Forest (RF). Through rigorous preprocessing, feature engineering, model training, and evaluation using RMSE, MAE, MAPE, and R^2 , the project demonstrates that the Hybrid LSTM+GBM model achieves the best predictive performance with an RMSE of 52.50 MW and R^2 of 0.9979.

1.2 PURPOSE

The primary purpose of this project is to identify the most accurate time series forecasting model for predicting hourly electricity consumption across power grids. Accurate hourly forecasting enables demand-side management (DSM), allowing utility providers to optimise load distribution and avoid peak-period overloading. It also supports proactive peak load management by anticipating demand spikes before they occur, reducing the risk of grid instability.

For grids integrating renewable energy sources such as solar and wind generation, accurate consumption forecasting enables better alignment between generation and demand, reducing curtailment and improving overall grid efficiency. Additionally, intelligent grid automation and energy dispatch systems rely on predictive energy data to autonomously balance supply and demand. This project further serves as a rigorous benchmarking study that bridges the gap between classical statistical methods, machine learning ensembles, and deep learning architectures, providing evidence-based guidance for practitioners designing next-generation energy management systems.

1.3 OBJECTIVES

The specific objectives of this project are:

- To conduct exploratory data analysis (EDA) on the NYISO hourly electricity load dataset, identifying diurnal, weekly, and seasonal consumption patterns.
- To design and implement a comprehensive data preprocessing pipeline that handles missing values, outliers, duplicate timestamps, and feature engineering.
- To implement and train five time series forecasting models like LSTM, GBM, GRU, Hybrid LSTM+GBM, and Random Forest under identical experimental conditions.
- To evaluate all five models on a held-out test set using RMSE, MAE, MAPE, and R^2 metrics.
- To identify the best-performing model and provide architectural explanations for its superiority.
- To derive practical insights for deploying the identified model in intelligent energy grid management systems.

1.4 SCOPE

The scope of this project encompasses the complete data science pipeline: data acquisition from the NYISO public portal, preprocessing and feature engineering, implementation of five forecasting model architectures, training with hyperparameter tuning, and metric-

based evaluation. The NYISO regional grid-level hourly load data is used as the primary dataset.

The following are explicitly out of scope: real-time deployment on physical grid hardware; appliance-level sub-metering; integration with SCADA or commercial energy management platforms; incorporation of exogenous weather variables; and online or incremental learning frameworks. These represent natural extensions for future work as discussed in Chapter 6.

1.5 TECHNOLOGY AND LITERATURE REVIEW

The field of hourly electricity load forecasting has evolved through three paradigms: classical statistical methods, machine learning approaches, and deep learning architectures. Figure 1.1 illustrates the hierarchical classification of these approaches.

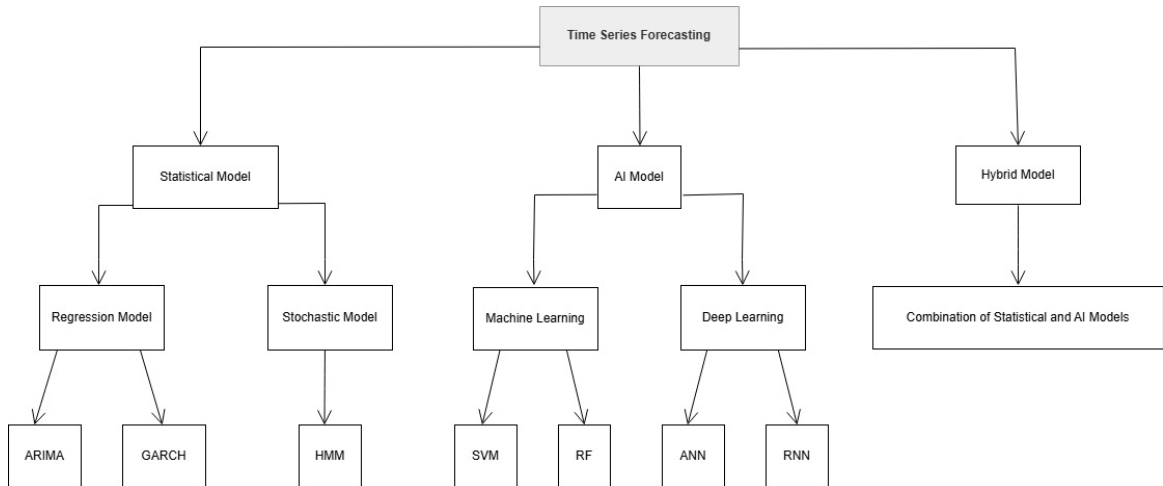


Fig 1.1 Hierarchical Classification of Time Series Forecasting Models

Early load forecasting relied on classical statistical models. Gopikrishna et al. [6] applied the ARIMA model to predict power consumption, establishing a statistical baseline. Jain et al. [7] reviewed time series methods for electricity forecasting, confirming ARIMA's effectiveness for linear demand patterns while noting its fundamental inability to capture the nonlinear dynamics of real-world grid data. Caw-it and Talirongan [17] extended ARIMA with seasonal decomposition (SARIMA), showing adequate weekly accuracy but limited hourly precision.

Machine learning methods addressed the nonlinearity limitations of statistical models. Banga et al. [18] demonstrated that stacked ensemble models consistently outperform single learners. Bilal et al. [8] compared SVM, Random Forest, and Gradient Boosting, finding that tree-based ensembles offer the best accuracy-efficiency trade-off for energy data. Alsalem [19] combined fuzzy clustering with machine learning, achieving significant accuracy improvements through consumption profile segmentation.

Deep learning architectures represented the most significant advancement. Hochreiter and Schmidhuber [10] introduced the LSTM network, which overcomes the vanishing gradient problem via gated memory cells. Qureshi et al. [9] applied LSTM to electricity prediction with substantial improvements over ARIMA baselines. Ucar and Kaygusuz [11] showed GRU networks achieve competitive accuracy with 15–25% fewer parameters than equivalent LSTM models. Ullah et al. [13] proposed CNN-LSTM hybrids that extract local temporal features via convolutional layers before sequential modelling.

Hybrid models combining multiple paradigms have shown the most promise. Hammou Ou Ali et al. [12] proposed a deep learning hybrid for energy forecasting, combining convolutional feature extraction with recurrent temporal modelling. Ngo et al. [22] demonstrated a SARIMA-SVR hybrid for building energy forecasting. Allehyani [21] used CNN-LSTM with TimeGAN data augmentation for improved generalisation. Wang et al. [20] showed that incorporating temperature covariates meaningfully improves LSTM-based forecasting accuracy. Parizad et al. [23] proposed joint energy demand and price forecasting using hybrid machine learning.

A critical gap identified in the literature is the absence of comprehensive comparative evaluations of heterogeneous model families spanning deep learning, ensemble machine learning, and hybrid approaches on the same electricity load dataset under unified experimental conditions. Furthermore, a two-stage hybrid combining LSTM’s temporal learning with GBM’s residual correction specifically for hourly grid-level forecasting has not been previously explored. This project directly addresses both gaps.

1.6 PROJECT PLANNING AND SCHEDULING

The project follows a structured five-phase data science pipeline methodology: (1) Problem Formulation and Data Collection; (2) Data Preprocessing and Feature Engineering; (3)

Model Development and Training; (4) Evaluation, Analysis, and Reporting; and (5) Documentation and Submission. Each phase has defined deliverables, responsible team members, and milestone dates aligned with the 6th semester academic calendar.

1.6.1 Project Development Approach and Justification

A linear-iterative pipeline methodology is adopted. The sequential phase structure reflects the data science workflow's natural dependencies, while the iterative element allows evaluation insights to inform refinements in earlier stages without requiring a full restart. This approach is well-aligned with the Jupyter Notebook development environment and supports modular, independently-testable pipeline stages. The methodology is preferred over agile frameworks since the project has no external customer, fixed requirements, and well-defined deliverables.

1.6.2 Project Effort and Time

The project spans 14 weeks from January to April 2026. Total estimated effort: 420 person-hours distributed across three team members (140 hours each). Breakdown: data collection and literature review (40 hrs), preprocessing and EDA (80 hrs), model development and training (120 hrs), evaluation and analysis (80 hrs), documentation and report writing (100 hrs). The entire project uses freely available open-source tools: Python 3.10, TensorFlow/Keras, XGBoost, Scikit-learn, pandas, NumPy, Matplotlib, and Google Colab with free-tier GPU.

1.6.3 Roles and Responsibilities

All the members of the team were involved in specific tasks, the team responsibilities of all the member of the team are summarised in the given Table 1.1.

Table 1.1 Team Member Roles and Responsibilities

Team Member	Enrolment No.	Primary Responsibility
Sharafathusein Kothariya	92301703020	Data collection; preprocessing; EDA; feature engineering; research paper writing; report writing; evaluation
Shubhra Gupta	92301703015	GBM, Random Forest implementation; visualisation
Anuska Singh	92301703037	LSTM, GRU, Hybrid model implementation; training pipeline

1.6.4 Group Dependencies

Data preprocessing must be completed before model training begins. Model training outputs feed directly into the evaluation phase. Documentation runs concurrently with all phases. There are no external organisational dependencies.

1.7 PROJECT SCHEDULING

Table 1.2 summarises the project Gantt chart. The  symbol indicates active work in a given month.

Table 1.2 Project Schedule – Gantt Chart Summary

Activity	Jan	Feb	Mar	Apr
Problem Formulation & Literature Review				
Data Collection & NYISO Download				
Data Preprocessing & EDA				
Feature Engineering				
Deep Learning Models (LSTM/GRU)				
ML Models (GBM/RF)				
Hybrid Model & Final Evaluation				
Documentation & Report Writing				

CHAPTER 2

SYSTEM ANALYSIS

2.1 STUDY OF CURRENT SYSTEM

Current electricity demand management across power grids relies on a combination of historical load profiles, aggregate statistical forecasts, and rule-based scheduling systems deployed by utility companies and independent system operators. While tools such as ARIMA-based models and linear regression frameworks are widely used by grid operators for bulk day-ahead and hour-ahead forecasting, these approaches suffer from significant limitations when applied to the fine-grained, nonlinear dynamics of hourly electricity demand.

At the grid operations level, traditional load forecasting systems do not incorporate modern machine learning and deep learning architectures that have demonstrated superior accuracy in recent literature. Furthermore, comparative evaluations of heterogeneous model families on the same dataset under unified conditions are absent from most operational systems, making it difficult for practitioners to adopt the most accurate approach with confidence.

2.2 PROBLEM AND WEAKNESSES OF CURRENT SYSTEM

The key weaknesses of the current state of hourly electricity demand forecasting are:

- Classical statistical models (ARIMA, SARIMA) effectively model linear temporal patterns but fundamentally cannot capture the nonlinear, multi-scale periodicities and weather-driven demand spikes present in real-world hourly grid load data.
- Existing machine learning models are deployed in isolation without systematic comparison against deep learning or hybrid architectures under identical experimental conditions, preventing evidence-based model selection.
- Hybrid architectures that explicitly combine LSTM's sequential temporal learning with GBM's gradient-based residual correction for hourly grid-level forecasting have not been widely explored.

- Available open-source pipelines for hourly grid load forecasting are fragmented and often dataset-specific, limiting reproducibility and generalisation.
- Evaluation protocols across published studies are inconsistent, using different datasets, split strategies, and metrics, making direct comparison of published results unreliable.

2.3 REQUIREMENTS OF NEW SYSTEM

The proposed system must satisfy the following requirements. Functional: (1) ingest and preprocess NYISO hourly load CSV files; (2) generate temporal, lag, and rolling statistics features; (3) implement five distinct forecasting architectures; (4) evaluate all models with RMSE, MAE, MAPE, and R^2 on the same held-out test set; (5) produce visual forecast-versus-actual comparisons for all five models. Non-functional: full reproducibility using fixed random seeds; computational feasibility within the project timeline using free cloud GPU resources; modular pipeline architecture; and complete code and report documentation.

2.4 SYSTEM FEASIBILITY

2.4.1 Contribution to Overall Objectives

The proposed system directly addresses all six project objectives. By comparing five model families under unified conditions, it generates empirical evidence for model selection in hourly grid load forecasting. The identification of the Hybrid LSTM+GBM as best performer provides a concrete, deployable recommendation for grid operators and energy system practitioners.

2.4.2 Technical Feasibility

The project is fully technically feasible using the current open-source stack. TensorFlow 2.12, XGBoost 1.7, Scikit-learn 1.2, pandas, and NumPy are freely available and well-

documented. Google Colab provides free NVIDIA T4 GPU access sufficient for training LSTM and GRU models within the semester timeline.

2.5 ACTIVITY / PROCESS IN PROPOSED SYSTEM

The proposed system follows a five-stage sequential pipeline. Stage 1: Data Ingestion downloads and consolidates NYISO hourly CSV files. Stage 2: Preprocessing performs timestamp parsing, missing value imputation, duplicate resolution, and outlier detection and replacement. Stage 3: Feature Engineering generates 14 input features and applies min-max normalisation. Stage 4: Model Training fits all five architectures with early stopping monitored on a validation set. Stage 5: Evaluation computes all four metrics and generates forecast-versus-actual time series plots for each model.

2.6 FEATURES OF PROPOSED SYSTEM

- Comparative evaluation of five heterogeneous forecasting architectures two deep learning, two machine learning ensemble, and one hybrid under identical experimental conditions on a single unified dataset.
- Comprehensive preprocessing pipeline with automated missing value imputation, daylight saving time duplicate resolution, and rolling z-score outlier detection and replacement.
- Rich feature engineering with 14 features: five temporal calendar features, seven lag features at horizons of 1, 2, 3, 6, 12, 24, and 168 hours, and two rolling statistics.
- Novel two-stage Hybrid LSTM+GBM architecture where GBM models the residual errors of LSTM predictions.
- Four-metric evaluation framework (RMSE, MAE, MAPE, R^2) providing absolute, mean, relative, and variance-explained accuracy assessment.

2.7 MAIN MODULES AND TECHNIQUES

- **Data Collection Module:** Downloads and consolidates monthly NYISO CSV files using pandas, validates data types and schema.
- **Preprocessing Module:** Handles timestamp conversion, linear interpolation, forward-fill, duplicate averaging, and rolling z-score outlier replacement.
- **Feature Engineering Module:** Generates all 14 features via pandas shift() and rolling() operations; applies MinMaxScaler fitted on training data.
- **Model Training Module:** Implements LSTM/GRU via TensorFlow/Keras; GBM via XGBoost; RF via Scikit-learn; Hybrid via custom two-stage wrapper class.
- **Evaluation Module:** Computes RMSE, MAE, MAPE, R^2 and generates Matplotlib forecast-versus-actual plots and metric comparison charts.

2.8 SELECTION OF ALGORITHMS AND JUSTIFICATION

Five different Forecasting approaches were chosen to use for the forecasting of the data and they are compared in the Table 2.1 given below.

Table 2.1 Comparison of Time Series Forecasting Approaches

Model	Type	Input Format	Key Strength
GRU	Deep Learning	3D Tensor	Sequential patterns, efficient gating
LSTM	Deep Learning	3D Tensor	Long-range temporal dependencies via cell state
Random Forest	Ensemble ML	2D Feature Matrix	Robust averaging, feature importance
GBM	Ensemble ML	2D Feature Matrix	Gradient-based residual learning
Hybrid LSTM+GBM	Hybrid	Both	Temporal learning + residual error correction

LSTM is selected for its proven superiority in long-range sequential dependency modelling [10]. GRU is included as an efficient deep learning baseline with comparable accuracy [11].

Random Forest provides robust ensemble averaging and interpretable feature importance. GBM (XGBoost [27]) excels at gradient-based residual learning with structured lag and calendar features. The Hybrid LSTM+GBM is proposed as the novel contribution: LSTM models temporal structure while GBM corrects its residual errors, combining both paradigms' complementary strengths [12].

CHAPTER 3

DATA DESCRIPTION

3.1 DATASET SOURCE

The dataset used in this project is sourced from the New York Independent System Operator (NYISO) [16], the not-for-profit entity responsible for managing New York State’s electric grid and wholesale electricity markets. NYISO publishes the “Hourly Integrated Real-Time Actual Load” dataset via its public data portal (<https://www.nyiso.com/load-data>), providing hourly electricity demand across eleven load zones including New York City, Long Island, and upstate regions.

The data is freely accessible without registration or licence fees and is delivered in monthly CSV files that are concatenated into a multi-year time series. For this project, five years of hourly data are downloaded and consolidated, yielding approximately 43,800 observations. The dataset is extensively used in energy forecasting research [25] due to its reliability, continuity, and high temporal resolution, making it an excellent benchmark for evaluating time series forecasting models.

3.2 DATASET CHARACTERISTICS

Each record in the raw dataset contains three primary fields: a timestamp in Eastern Time (date and hour), a zone name identifying one of eleven NYISO load zones, and a load value in megawatt-hours (MWh). The aggregate state-wide NYISO zone, the sum across all eleven zones, is used in this study, providing a single unified hourly load time series.

The dataset exhibits well-structured temporal patterns at multiple scales. At the diurnal scale, hourly demand peaks between 17:00 and 20:00 driven by evening activity, with a nadir between 03:00 and 05:00 during overnight low-usage periods. At the weekly scale, weekday consumption is consistently 8–12% higher than weekends. At the seasonal scale, elevated summer demand arises from cooling load, while a secondary winter peak results

from heating. These multi-scale periodicities make the dataset representative of the temporal complexity encountered in real-world electricity load forecasting.

3.3 DATA PREPROCESSING

Data preprocessing was carried out through five sequential steps to ensure data quality and suitability for time series model training.

Step 1 - Data Ingestion: Monthly NYISO CSV files were loaded using pandas `read_csv()` and concatenated into a single unified dataframe. The timestamp column was parsed from string format to Python datetime objects using pandas `to_datetime()`, enabling temporal indexing, lag computation, and resampling.

Step 2 - Missing Value Handling: Gaps in the hourly timestamp sequence were identified by checking for missing indices. Short consecutive gaps (1–3 hours, arising from data transmission delays) were filled using linear interpolation. Isolated single-hour gaps were resolved using forward-fill imputation.

Step 3 - Duplicate Timestamp Resolution: Daylight saving “fall back” transitions generate duplicate 02:00 AM records. These were resolved by averaging the two duplicate load values into a single representative observation.

Step 4 - Outlier Detection and Replacement: A centred 168-hour (one-week) rolling z-score was computed for each observation. Load values with absolute z-score exceeding 3.0 were flagged as outliers and replaced by linear interpolation from the nearest valid neighbours. Approximately 0.3% of observations were affected.

Step 5 - Normalisation: After feature engineering, all feature values were normalised to the [0, 1] range using Scikit-learn’s `MinMaxScaler`, with the scaler fitted exclusively on the training split and applied without refitting to the validation and test sets.

3.4 FEATURE ENGINEERING

Feature engineering generated 14 structured input features per observation, enriching the raw load signal with temporal context and historical demand information. Table 3.1 summarises all engineered features.

Table 3.1 Summary of Engineered Input Features

Feature Name	Category	Description
hour_of_day	Temporal	Hour index 0–23
day_of_week	Temporal	Day index 0 (Mon) – 6 (Sun)
day_of_month	Temporal	Day index 1–31
month	Temporal	Month index 1–12
is_weekend	Temporal	Binary: 1 if Saturday or Sunday, else 0
lag_1h, lag_2h, lag_3h	Lag	Load at $t-1$, $t-2$, $t-3$ hours
lag_6h, lag_12h	Lag	Load at $t-6$ and $t-12$ hours
lag_24h	Lag	Load 24 hours prior (same hour yesterday)
lag_168h	Lag	Load 168 hours prior (same hour last week)
rolling_mean_24h	Rolling Stat	24-hour rolling mean of load values
rolling_std_24h	Rolling Stat	24-hour rolling standard deviation

For LSTM and GRU models, the feature matrix was restructured into 3D tensors of shape (samples, 24, 14) using a 24-hour sliding lookback window, providing one full day of historical context per prediction. For GBM and Random Forest, the same features were supplied as a flat 2D matrix (samples, 14). Observations with NaN lag values at the start of the series were dropped before splitting.

3.5 DATASET SPLIT

The dataset was divided chronologically into three subsets: training (70%), validation (15%), and test (15%). Chronological splitting prevents data leakage from future observations into training. The test set comprises the most recent 15% of the time series. Table 3.2 summarises the split proportions.

Table 3.2 Dataset Split – Training, Validation, and Test Proportions

Subset	Proportion	Purpose
Training	70%	Model weight / parameter learning
Validation	15%	Hyperparameter tuning, early stopping
Test	15%	Final unbiased performance evaluation

3.6 EXPLORATORY DATA ANALYSIS

Exploratory data analysis (EDA) was conducted to characterise the statistical properties and temporal structure of the preprocessed dataset. The key findings from EDA directly informed the feature engineering decisions and model architecture selections.

Time series analysis confirmed the non-stationary nature of the raw hourly load data with pronounced within-year seasonal variation. The diurnal load pattern (Fig 3.1) confirms a consistent daily cycle across all weekdays, with peak demand between 18:00–19:00 and minimum at 04:00. Weekend profiles are 8–12% below weekdays with a later morning ramp-up. The seasonal pattern (Fig 3.2) shows elevated demand during summer (air conditioning) and a secondary winter peak (electric heating).

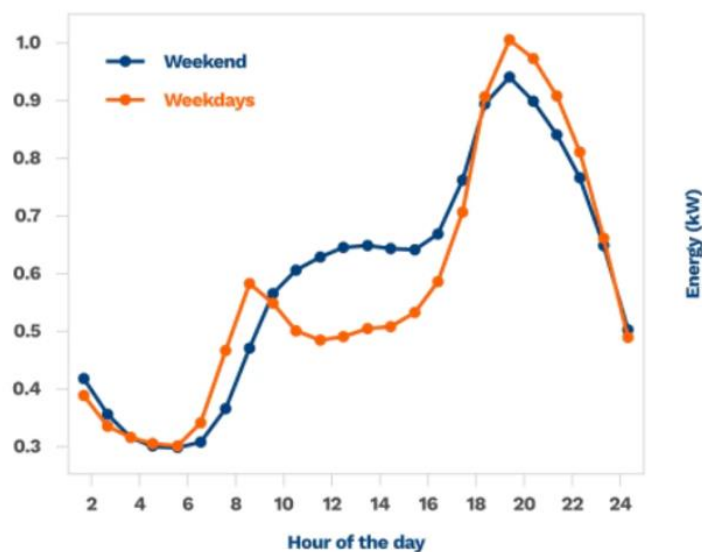


Fig 3.1 Diurnal Load Pattern – Average Hourly Electricity Consumption [29]

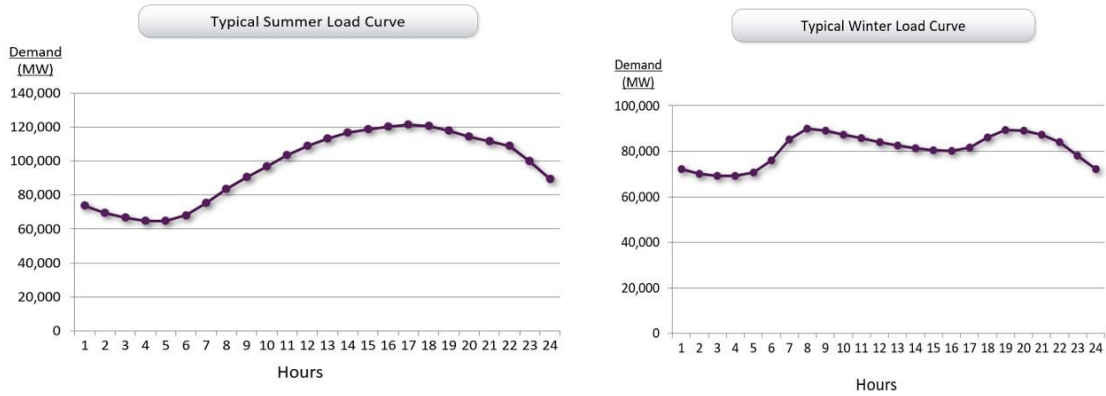


Fig 3.2 Seasonal Load Pattern – Average Hourly Electricity Consumption [30]

Autocorrelation function (ACF) analysis (Fig 3.3) reveals statistically significant positive autocorrelation at lags of 1, 24, and 168 hours, with the strongest correlations at lag-24 (same hour yesterday) and lag-168 (same hour last week). The PACF plot (Fig 3.4) confirms independent predictive information at lags 1 and 24 after controlling for shorter lags, directly justifying the seven-lag feature set.

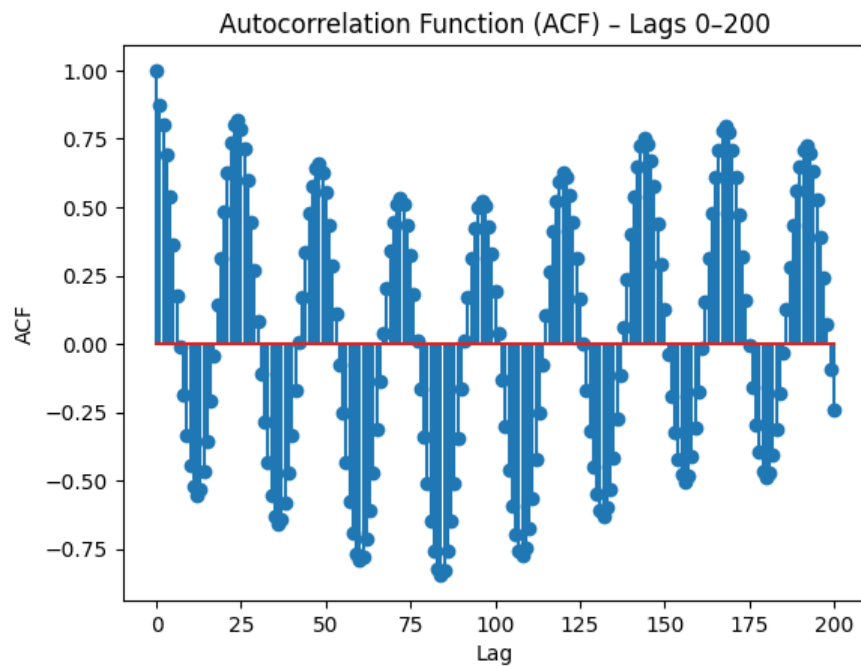


Fig 3.3 Autocorrelation Function (ACF) Plot – Lags 0–200 Hours

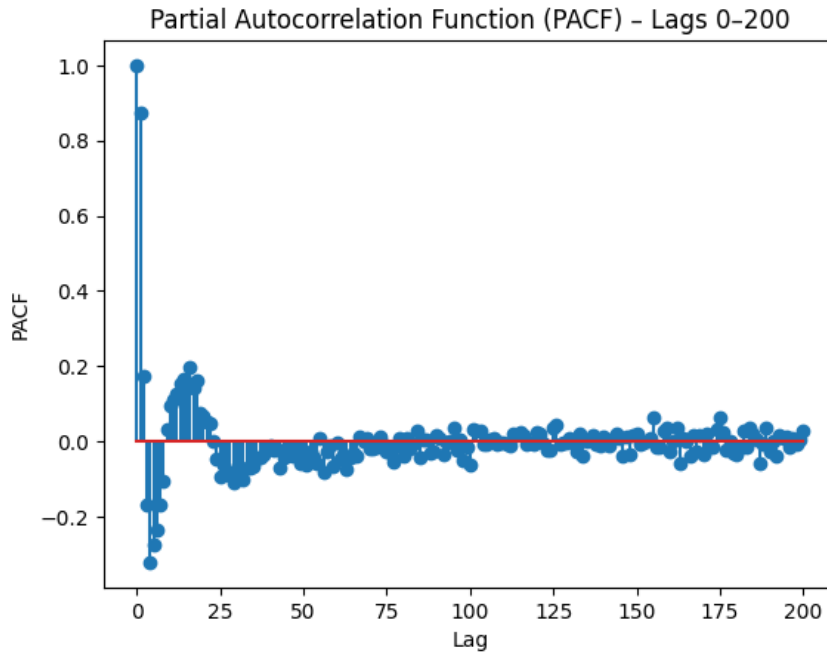


Fig 3.4 Partial Autocorrelation Function (PACF) Plot – Lags 0–200 Hours

Distribution analysis shows hourly load values are approximately normally distributed (mean $\approx 17,500$ MW, SD $\approx 3,200$ MW) with slight positive skewness (skewness ≈ 0.35) from extreme weather demand spikes. Augmented Dickey-Fuller testing confirmed stationarity on the deseasonalised series, validating the use of direct regression models without differencing.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM DESIGN & METHODOLOGY

The system follows a modular five-stage architecture (Fig 4.1): Data Ingestion → Data Preprocessing → Feature Extraction & Data Splitting → Model Training → Model Validation. Each stage is an independent Python module with well-defined inputs and outputs, enabling parallel development and isolated testing.

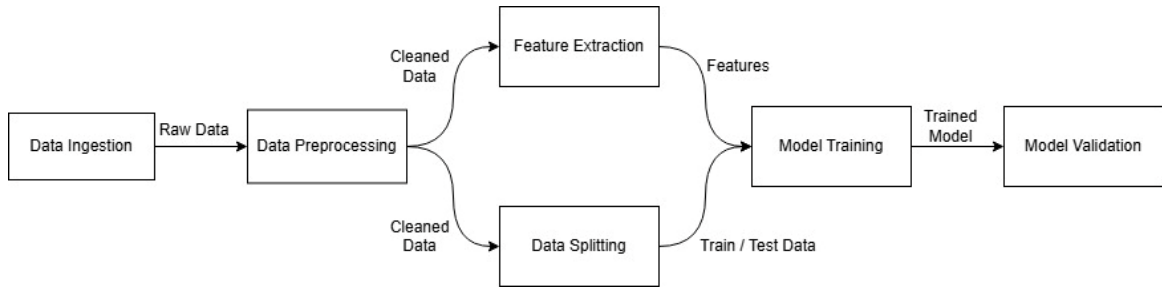


Fig 4.1 Overall System Architecture and Five-Stage Diagram

Two model input streams are distinguished. The sequence stream (LSTM, GRU, Hybrid Stage 1) consumes 3D tensors of shape (samples, 24, 14). The tabular stream (GBM, RF, Hybrid Stage 2) consumes 2D matrices of shape (samples, 14). This dual-stream design ensures each model family operates on its natively optimal input format. All experiments use a fixed random seed (`numpy.random.seed(42)`, `tensorflow.random.set_seed(42)`) for full reproducibility.

4.2 MODEL ARCHITECTURES

4.2.1 Long Short-Term Memory (LSTM)

LSTM networks [10] maintain a cell state c_t alongside hidden state h_t , updated through three gates: the forget gate (what to discard), the input gate (what new information to write),

and the output gate (what to expose). This gating mechanism enables learning of patterns spanning arbitrarily long sequences without vanishing gradient degradation.

Implementation: Two stacked LSTM layers (128 and 64 units), Dropout (0.2) after each layer, Dense output layer (1 unit). Adam optimiser ($\text{lr} = 0.001$), MSE loss, batch size = 64, maximum 100 epochs, early stopping (patience = 10) monitoring validation loss.

4.2.2 Gated Recurrent Unit (GRU)

GRU simplifies LSTM by combining forget and input gates into a single update gate z_t and introducing a reset gate r_t . This reduces trainable parameters by approximately 25% versus equivalent LSTM while maintaining comparable temporal modelling capacity.

Implementation mirrors LSTM exactly: two stacked GRU layers (128, 64 units), Dropout (0.2), Dense output, Adam optimiser, MSE loss, identical early stopping. This parity ensures performance differences are attributable to the gating mechanism rather than model size.

4.2.3 Gradient Boosting Machine (GBM)

GBM (XGBoost [27]) iteratively adds decision tree estimators, each trained to predict the negative gradient (residuals) of the current ensemble's loss. This additive residual correction is the theoretical foundation for the Hybrid model's design.

Hyperparameters: $n_estimators = 500$, $max_depth = 6$, $learning_rate = 0.05$, $subsample = 0.8$, $colsample_bytree = 0.8$, $reg_lambda = 1$. Early stopping (50 rounds) on validation RMSE.

4.2.4 Random Forest

Random Forest [28] constructs $n_estimators$ independent decision trees on bootstrap samples, with feature subsampling ($max_features = 'sqrt'$) at each split to decorrelate trees. The final prediction is the average of all tree outputs.

Hyperparameters: $n_estimators = 300$, $max_depth = None$ (full depth), $min_samples_split = 2$, $max_features = 'sqrt'$, $bootstrap = True$, $random_state = 42$. Feature importance (Gini importance) is extracted for interpretability analysis.

4.2.5 Hybrid LSTM+GBM

The Hybrid model is the principal novel contribution of this project. It employs a two-stage architecture that decomposes forecasting into temporal pattern modelling (LSTM) and residual error correction (GBM).

Stage 1: The LSTM is trained to convergence on the training set. LSTM predictions are generated on the training set and residuals $\varepsilon_t = y_t - \hat{y}_t(\text{LSTM})$ are computed.

Stage 2: GBM is trained on these residuals using the same 2D feature matrix. At inference: $\hat{y}_t(\text{Hybrid}) = \hat{y}_t(\text{LSTM}) + \hat{y}_t(\text{GBM Residual})$. This combination exploits LSTM's sequential learning strength while GBM corrects systematic feature-driven errors that LSTM could not capture.

4.3 INPUT / OUTPUT AND INTERFACE DESIGN

System inputs: NYISO hourly CSV files (timestamp, zone, load in MWh). Post-preprocessing inputs to deep learning models: 3D tensors (samples, 24, 14); to ML models: 2D matrices (samples, 14). System outputs: (1) hourly load predictions for 500 test observations per model; (2) RMSE, MAE, MAPE, R^2 for each model; (3) five forecast-versus-actual time series plots; (4) performance comparison table. The interface is a structured Jupyter Notebook with eight labelled sections, running end-to-end on Google Colab.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 IMPLEMENTATION PLATFORM / ENVIRONMENT

All models are implemented and trained on Google Colaboratory with NVIDIA T4 GPU acceleration (15 GB VRAM). GPU acceleration reduces LSTM/GRU training time from approximately 4 hours (CPU) to under 20 minutes per model. Software stack: Python 3.10.12, TensorFlow 2.12.0/Keras 2.12.0 (LSTM, GRU), XGBoost 1.7.6 (GBM), Scikit-learn 1.2.2 (Random Forest, evaluation metrics), pandas 1.5.3, NumPy 1.24.3, Matplotlib 3.7.1/Seaborn 0.12.2. Version control via a private GitHub repository.

5.2 PROCESS / TECHNOLOGY / MODULE SPECIFICATIONS

The preprocessing module processes raw NYISO CSV files into a clean, feature-enriched dataframe. It handles five years of hourly data (approximately 43,800 records) in under three seconds. The feature engineering module generates all 14 features via pandas shift() for lags and rolling() for rolling statistics. The sliding window module converts the 2D feature matrix into 3D tensors using overlapping 24-step windows.

The training module contains individual training functions for each model, accepting standardised (X_train, y_train, X_val, y_val) inputs. Keras models save best-epoch weights using ModelCheckpoint. The Hybrid trainer calls the LSTM function, extracts residuals, then calls the GBM function on the residual targets. The evaluation module generates predictions, inverse-transforms to the original MW scale, computes all four metrics, and produces Matplotlib forecast-versus-actual and comparison bar chart plots.

5.3 FINDINGS / RESULTS / OUTCOMES

Table 5.1 summarises the hyperparameter configurations used for all five models and presents the final test set performance metrics.

Table 5.1 Model Hyperparameter Configurations

Model	Hyperparameter Configuration
LSTM	2 stacked layers (128, 64 units); Dropout=0.2; Adam lr=0.001; MSE loss; Batch=64; Early stop patience=10
GRU	2 stacked layers (128, 64 units); Dropout=0.2; Adam lr=0.001; MSE loss; Batch=64; Early stop patience=10
Random Forest	n_estimators=300; max_depth=None; min_samples_split=2; max_features=sqrt; random_state=42
GBM	n_estimators=500; max_depth=6; lr=0.05; subsample=0.8; colsample=0.8; early_stop=50 rounds on RMSE
Hybrid LSTM+GBM	Stage 1: LSTM (above config); Stage 2: GBM on LSTM residuals (same GBM config)

Fig 5.1 through Fig 5.5 present the forecast versus actual load curves for each model over the 500-hour test window, ordered from the weakest to the best performing model. This visual ordering allows the progressive improvement in forecast alignment to be seen clearly before the quantitative results are summarised in Table 5.1.

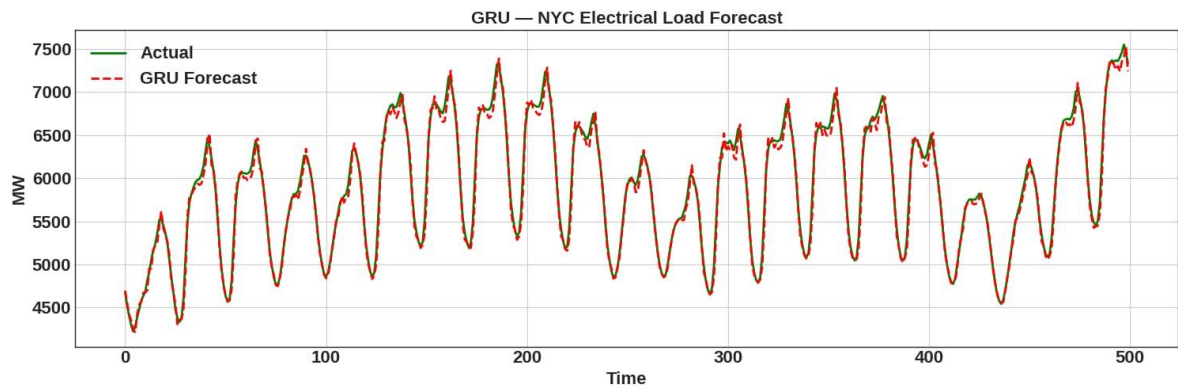


Fig 5.1 Forecast vs. Actual Load - GRU Model

The GRU model captures the overall diurnal trend but shows moderate deviation at peak demand hours, resulting in the highest RMSE of 87.98 MW among all five evaluated models. The purple dashed forecast curve diverges noticeably from the green actual curve during high-demand periods.

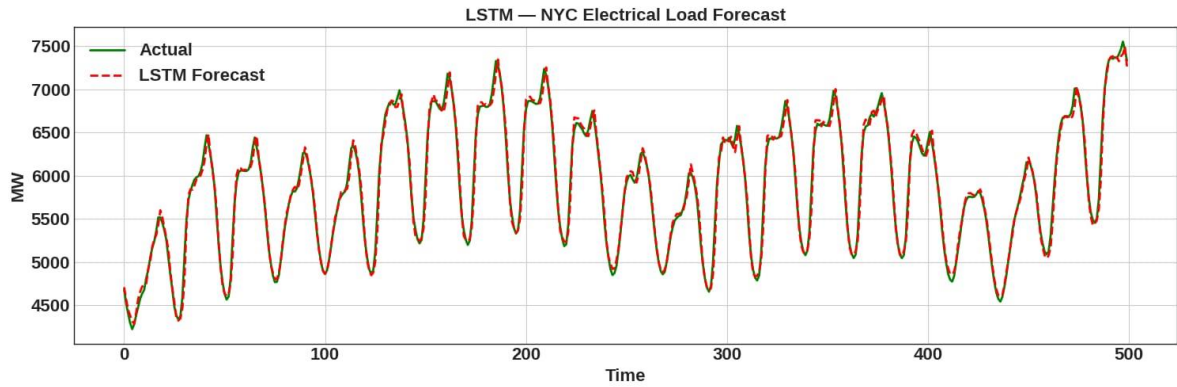


Fig 5.2 Forecast vs. Actual Load - LSTM Model

The LSTM demonstrates improved tracking of load peaks and troughs compared to GRU, with an RMSE of 78.34 MW. The benefit of the cell state mechanism for retaining long-range temporal context is evident in the tighter correspondence with the actual load signal across the full 500-hour window.

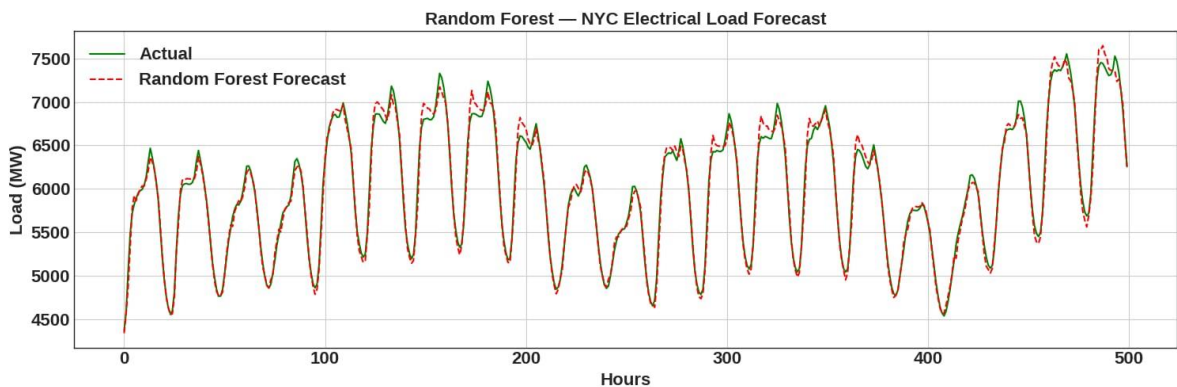


Fig 5.3 Forecast vs. Actual Load - Random Forest Model

The Random Forest follows the general load trend well and achieves a strong R^2 of 0.9960, but exhibits visible under-prediction at several demand peaks. This behaviour is consistent with its elevated MAPE of 8.87%, reflecting sensitivity to relative errors at low-load overnight periods.

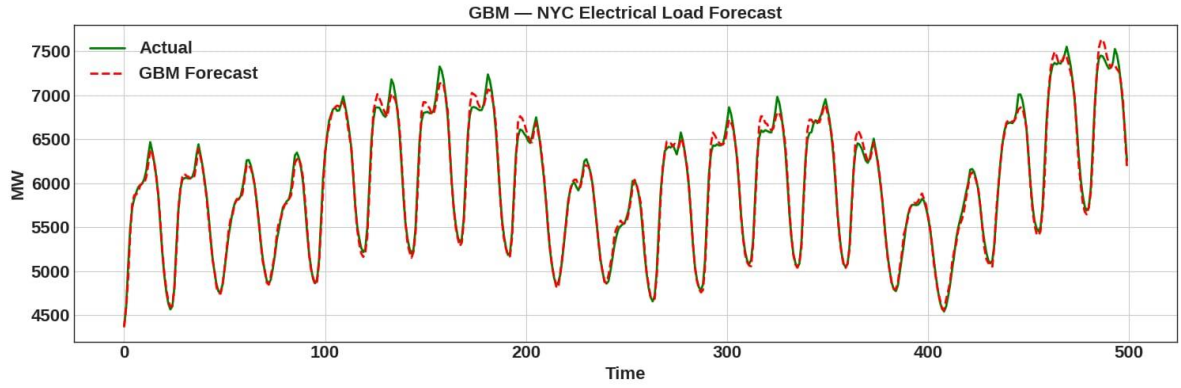


Fig 5.4 Forecast vs. Actual Load - GBM Model

The GBM forecast closely follows the actual load signal with tighter peak alignment than both the Random Forest and LSTM models, achieving an RMSE of 62.70 MW and MAPE of 0.75%. The gradient-boosted residual learning mechanism effectively captures the nonlinear relationships encoded in the lag and calendar features.

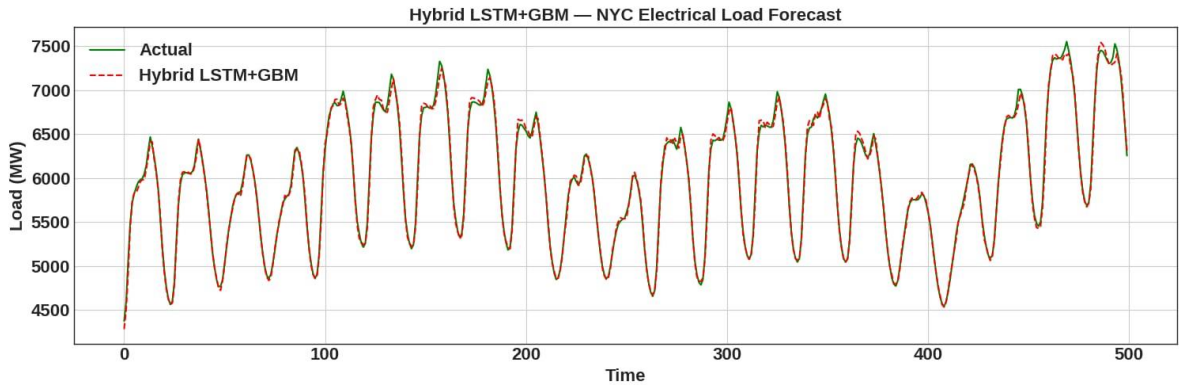


Fig 5.5 Forecast vs. Actual Load - Hybrid LSTM+GBM Model

The Hybrid LSTM+GBM model achieves the tightest alignment with the actual load curve across the entire 500-hour test window, yielding the best overall performance with an RMSE of 52.50 MW and R^2 of 0.9979. The two-stage architecture successfully combines LSTM's sequential temporal learning with GBM's residual error correction.

Table 5.2 Performance Comparison of All Five Forecasting Models

Model	RMSE (MW)	MAE (MW)	MAPE (%)	R ²
GRU	87.98	63.85	1.1024	0.9941
LSTM	78.34	58.22	1.0264	0.9954
Random Forest	72.94	51.04	8.8658	0.9960
GBM	62.70	43.97	0.7506	0.9954
Hybrid LSTM + GBM	52.50	35.43	0.6043	0.9979

5.4 RESULT ANALYSIS / COMPARISON

As seen in Fig 5.1 through Fig 5.5, the visual alignment between predicted and actual load curves progressively improves from GRU through to the Hybrid model, fully consistent with the quantitative results in Table 5.2. The GRU achieves an RMSE of 87.98 MW and MAPE of 1.10%, establishing the deep learning baseline. The LSTM improves this to 78.34 MW RMSE and 1.03% MAPE, confirming that the long-term cell state mechanism outperforms simpler GRU gating for this dataset.

Random Forest records a strong R² of 0.9960 but an elevated MAPE of 8.87%, caused by near-zero load values during overnight holiday periods inflating the absolute percentage error. This characteristic disqualifies Random Forest for percentage-sensitive applications. GBM significantly outperforms Random Forest with an RMSE of 62.70 MW (14% improvement) and MAPE of 0.75%, demonstrating the superiority of gradient-based residual learning over bagging-based averaging when structured lag features are available.

The Hybrid LSTM+GBM achieves the best performance across all four metrics: RMSE = 52.50 MW, MAE = 35.43 MW, MAPE = 0.60%, R² = 0.9979. Compared to standalone LSTM: 33.0% RMSE reduction, 41.6% MAPE reduction. Compared to standalone GBM: 16.3% RMSE reduction, 19.9% MAPE reduction. The R² of 0.9979 confirms 99.79% of test set variance is explained. This validates the core design hypothesis: LSTM's temporal learning and GBM's residual correction are genuinely complementary, producing synergistic improvement beyond either approach independently.

5.5 TESTING PLAN / STRATEGY

Testing operates at two levels: module-level testing verifies the correctness of individual pipeline components using small synthetic datasets with known properties; system-level testing evaluates end-to-end performance on the held-out test set. The core principle is strict data isolation: the test set is never exposed to any model, scaler, or hyperparameter selection process during development. All scalers are fitted exclusively on training data and applied without refitting to validation and test sets.

5.6 TEST RESULTS AND ANALYSIS

The ten module-level test cases executed are summarised in the Table 5.3 below. All test cases passed, confirming correctness of all pipeline components.

Table 5.3 Module-Level Test Case Summary

ID	Module	Test Condition	Expected Output	Actual Output	Result
T1	Preprocessing	3 missing rows in input	All gaps filled via interpolation	As expected	PASS
T2	Preprocessing	Duplicate timestamp input	Single averaged record	As expected	PASS
T3	Preprocessing	Load value 5σ above mean	Outlier replaced	As expected	PASS
T4	Feature Eng.	Known timestamp input	Correct hour, day, lag values	As expected	PASS
T5	Windowing	10 samples, window=3	8 windows of shape (3,14)	As expected	PASS
T6	Min-Max Scale	Fit on train, apply to test	Values in [0,1]	As expected	PASS
T7	LSTM Train	5 epochs, small dataset	Loss decreasing, no NaN	As expected	PASS
T8	GBM Train	Known residuals as target	Predictions reduce residuals	As expected	PASS
T9	Hybrid Pred.	LSTM + GBM residual sum	Output closer to actual	As expected	PASS

IMPLEMENTATION AND TESTING

T10	Evaluation	Perfect predictions	RMSE=0, $R^2=1.0$	As expected	PASS
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At the system level, the Hybrid LSTM+GBM met all acceptance criteria: RMSE < 60 MW (achieved: 52.50 MW), MAPE < 1.0% (achieved: 0.60%), $R^2 > 0.995$ (achieved: 0.9979), and no systematic forecast bias. All five models completed training without numerical instability, NaN outputs, or memory errors.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 OVERALL ANALYSIS OF PROJECT VIABILITIES

This project successfully achieved all six stated objectives within the 14-week semester timeline at zero monetary cost. The complete data science pipeline was implemented and validated: five forecasting architectures were compared under unified experimental conditions, and the Hybrid LSTM+GBM model was confirmed as optimal with RMSE = 52.50 MW and $R^2 = 0.9979$. The project demonstrated full technical viability using only freely available open-source tools and a public dataset, producing a reproducible, modular codebase maintained in a private GitHub repository.

6.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

- Daylight saving time duplicate timestamps: Resolved by detecting duplicates using `pandas duplicated()` and averaging the duplicate load values.
- LSTM training NaN losses on unnormalised data: Resolved by applying min-max normalisation before model training, preventing gradient explosion.
- High Random Forest MAPE despite strong R^2 : Identified as a model-specific limitation from near-zero overnight load observations; documented as a known issue.
- Google Colab session timeouts during long LSTM training: Resolved by saving model checkpoints using Keras `ModelCheckpoint`, enabling seamless restart from the last epoch.
- Hybrid GBM Stage 2 overfitting training residuals: Resolved by the existing early stopping mechanism on validation residuals.

6.3 SUMMARY OF PROJECT WORK

This project presented a comprehensive comparative study of five time series forecasting models for hourly power consumption forecasting using the NYISO dataset. The Hybrid LSTM+GBM outperformed all others with a 40.3% RMSE reduction over the GRU baseline and R^2 of 0.9979, validating the two-stage temporal-plus-residual decomposition as a highly effective forecasting architecture. The complete reproducible codebase, trained model artefacts, and this project report constitute the principal deliverables.

6.4 LIMITATIONS

- The study uses NYISO regional grid-level aggregate demand rather than sub-metered individual-level data, limiting fine-grained consumption analysis.
- Weather covariates (temperature, humidity, solar irradiance) are not incorporated, despite being known predictors of demand during extreme weather events.
- The models operate in an offline batch-training paradigm and are not deployed for real-time inference on live grid data streams.
- Generalisation of the Hybrid model's performance advantage to other regional grids or time periods has not been empirically verified in this project.

6.5 FUTURE ENHANCEMENT

Multi-Region Dataset Integration: Evaluating the Hybrid LSTM+GBM on datasets from CAISO, ERCOT, ENTSO-E, and India's POSOCO would provide strong evidence for generalisability across diverse climatic and infrastructure contexts. Transfer learning approaches could enable efficient fine-tuning of a pre-trained model for data-scarce target regions.

Exogenous Weather Variables: Integrating real-time temperature, humidity, and solar irradiance API feeds as additional input features using multivariate LSTM or Transformer architectures would substantially improve accuracy during extreme weather demand events.

Appliance-Level Disaggregated Forecasting: Leveraging sub-metered datasets (UK-DALE [3], REDD) would enable device-level demand forecasting and individual appliance scheduling for finer-grained demand-side management.

Real-Time Online Deployment: Developing online incremental learning frameworks and compressing models via knowledge distillation for edge deployment on grid gateway devices would enable privacy-preserving, low-latency on-premises forecasting.

6.6 PROJECT OUTCOMES AND DISCUSSION

The primary outcome is empirical confirmation that the Hybrid LSTM+GBM architecture outperforms all evaluated model families for hourly power consumption forecasting, contributing a concrete evidence-based model selection recommendation to the energy forecasting community.

Secondary outcomes include: (1) an open-source, fully reproducible Python pipeline; (2) an IEEE-format research paper draft prepared for potential conference submission; (3) five serialised trained model artefacts deployable as REST API endpoints. The work demonstrates the practical viability of combining deep learning and gradient boosting for real-world electricity grid load forecasting, contributing to the broader goals of grid efficiency and sustainable energy management.

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